Subset Sums in a Fourfold Sumset

A checked continuation from the upper bound 115 to 103

Sum-of-Subset research project

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Abstract

Let $\mathrm{FS}(A)$ include the empty subset sum, and let $4A$ allow repeated summands. Starting with the previously audited bound $N \le 115$ for $\mathrm{FS}(A) \subseteq 4A$, we exclude every cardinality from 104 through 115 by one common argument. The resulting independently checked continuation gives $71 \le N \le 103$. Its finite part consists of 24 explicit hole witnesses and exact scalar inequalities, verified by a short standard-library Python program. It uses no numerical optimization or search over candidate sets. The structural lemmas are proved here, with the audited 115 foundation explicitly identified. The repository's earlier claim $N \le 84$ is stronger; this report does not claim a new project record or certify that entire chain.

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1 Scope, prerequisites, and the main conclusion

For a finite real set X, put

$$\mathrm{FS}(X) = \left\{ \sum_{x \in Y} x : Y \subseteq X \right\}, \qquad R(X) = \{x + y : x, y \in X,\ x \neq y\}, \quad \Delta(X) = |2X| - |R(X)|.$$

Here $2X = X + X$, $T_r = r(r+1)/2$, and N is the largest feasible cardinality. The following theorem concerns all real sets, not only integer sets or sets of a prescribed shape.

Theorem 1.1. *If* $\mathrm{FS}(A) \subseteq 4A$, *then* $|A| \le 103$.

The 71-element construction from the accompanying research report remains unchanged:

$$A = [-31, 31]_{\mathbb{Z}} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}.$$

The precise maximum is not determined here.

1.1 The audited starting point

This is a continuation of the 115 foundation, not a replacement for its proof. We use the following previously checked facts, as stated in Part I of `upper86_complete_20261005.tex`, with the peeling correction in `proof_review_20261014.md`.

1. A feasible set remains feasible on adjoining zero. Any feasible real set has an integer realization of the same cardinality and sign order, preserving zero when present. To see the latter, fix one four-term representation for every subset sum. These are homogeneous rational linear equations; rational points are dense in their real kernel. Preserve the strict order inequalities and then clear denominators.

2. Every feasible set has at most 115 elements.

3. Remove the four largest positive and four most negative elements, taking all of a sign if fewer than four exist. With remainder D, $d = |D|$,

$$|\mathrm{FS}(D)| \le 2|2D| + 18d + 85. \tag{1}$$

   The refined count consists of two leading $2D$ translates, eighteen ordinary units of capacity d, at most 84 nonzero fixed contributions, and zero. Its 110 fixed labels and 26 disjoint nonpositive groups are also reproduced and checked in the newer research report and `verification/verify.py`.

4. A four-extreme peel, preserving zero, gives

$$|\mathrm{FS}(F)| \ge |\mathrm{FS}(F')| + |R(F)| - 1. \tag{2}$$

   When both signs suffice, remove two elements at each extreme. If the extreme deleted sums are α, β, the residue-class argument counts the classes of $\alpha + \mathrm{FS}(F')$ modulo $\beta - \alpha$, not the original classes of $\mathrm{FS}(F')$. In each translated class the two extreme translates provide one additional point. Restricted sums in other classes then prove (2). This is the correction used throughout this report.

5. For $X_i = \{0 < x_1 < \cdots < x_i\}$, write

$$\mu(X_i) = |\mathrm{FS}(X_i)| - 1 - T_i = \sum_{j=1}^{i} y_j, \qquad y_j \ge 0,$$

$$\delta_0(X_i) = |\{x \in X_i : 2x \notin R(X_i \cup \{0\})\}|, \quad \rho(X_i) = \mu(X_i) + 2 - 2\delta_0(X_i).$$

   Then $\Phi = 2 - \rho = 2\delta_0 - \mu \le 10$, and $\rho(X_i) \ge \rho(X_{i-1}) + y_i - 2$. If $i \ge 6$ and $y_i \le i - 5$, the small-increment inverse lemma gives

$$X_i = t([1, h]_{\mathbb{Z}} \setminus D_0), \quad i \le h \le i + y_i, \quad |D_0| = h - i, \quad h \notin D_0. \tag{3}$$

   The circle-group argument proving this lemma uses the subset-sum growth results of Balandraud–Girard–Griffiths–Hamidoune [2]. Equality $y_i = 0$ yields an initial progression; for $i \ge 6$, $y_i = 0, 1$ implies $\Phi(X_i) \le 2$.

We additionally use the following precise external results. The stability theorem of Carpenter–Defant–Kravitz [1, Theorem 1.1] says that for $r \geq 4$ and $0 \leq M \leq r-4$, $\mu(X) \leq M$ implies $X = tB$, where B is a positive integer set of size r and $\sum B \leq T_r + M$. In particular a non-progression subset-sum support has $\mu \geq r-3$. The two-set inverse theorem and central-interval theorem are stated at their actual applications below. We do not assume any post-115 endpoint exclusion from the archived reports.

2 Positive-side defect estimates

Lemma 2.1 (Deleted intervals)**.** *If $X = [1, r+q]_{\mathbb{Z}} \setminus D$, $|D| = q$, $r+q \notin D$, and $r \geq \max(5, 4q+1)$, then $\Phi(X) \leq 2$. For $q \geq 1$ one also has $\delta_0(X) \leq 2q$. The latter defect bound remains valid for $q = 1, 2$ and $r \geq 7$.*

Proof. The case $q = 0$ is immediate. Write $h = r+q$ and $C = \{h - d : d \in D\}$. The integer mass excess is

$$e_0 = \sum X - T_r = \sum_{c \in C} c - T_{q-1}.$$

There are $\min(k, h-k)$ pairwise disjoint representations of $2k$ by distinct members of $[0, h]_{\mathbb{Z}}$. Thus a defective present k must be in $[1, q]$ or $[r, h]$. At most $q+1$ high defects occur. For $q \geq 2$, all $1, \ldots, q$ cannot be both present and defective: defectivity of 1 forces 2 absent. Hence at most $q-1$ low defects occur. For $q = 1$, the only extra defect besides the maximum can arise from deleting 2 or $h-2$; these are distinct for $r \geq 5$. This proves the defect counts.

Since $r > q$, X is primitive. If $\mu \leq r-4$, stability implies $\mu = e_0$: every integer normalization of a primitive integer set is an integer dilation, while $|\operatorname{FS}(X)| \leq \sum X + 1$. With no low defects, let the high defects other than h be $h - t$. Then $t \notin C$ and $2t \in C$. Replacing those depths $2t$ by the absent halves t leaves q distinct positive integers. If there are v such defects,

$$e_0 \geq q + \sum t \geq q + v \geq 2v = 2\delta_0 - 2.$$

If there are $\ell \geq 1$ low defects, their forced holes $2k$ have depths at least $r - q$, spaced by at least two. Therefore

$$e_0 \geq \ell(r-q) + \ell(\ell-1) + T_{q-\ell} - T_{q-1}.$$

Subtracting $2q + 2\ell$ from the right side gives

$$\ell(r - 2q - 2) + \tfrac{3}{2}\ell(\ell-1) - q \geq 0$$

under $r \geq 4q+1$. Since $\delta_0 \leq q + 1 + \ell$, again $e_0 \geq 2\delta_0 - 2$. Finally, if $\mu > r-4$, then $\Phi \leq 4q - (r-3) \leq 2$. □

Lemma 2.2. *For a positive real set of size $r \geq 14$, $\rho \geq 0$. If $r \geq 6$ and its subset-sum support is not an arithmetic progression, then $\rho \geq r - 14$.*

Proof. For $9 \leq i \leq 14$, put $q_i = \lfloor (i-1)/4 \rfloor$. If $y_i \leq q_i$, (3) and Lemma 2.1 give $\Phi_i \leq 2$. Otherwise $\Phi_i \leq \Phi_{i-1} + 1 - q_i$. Starting at $\Phi_8 \leq 10$ gives successive bounds $9, 8, 7, 6, 4, 2$. Larger increments $y_i \geq 2$ preserve the bound, and the audited cases $y_i = 0, 1$ do also.

For the second assertion, start with $\rho \geq -8$ at $r = 6$ and induct. If $y_r \leq 2$, (3) gives at most two deletions, hence $\delta_0 \leq 4$, while stability gives $\mu \geq r-3$. Thus $\rho \geq r-9$. If $y_r \geq 3$ and the preceding support is not a progression, the defect recurrence gives the result. If the preceding support is $t[0, L]_{\mathbb{Z}}$, its next translate must be disjoint: any overlap would again produce a progression. Consequently $y_r \geq (r-1)(r-2)/2$, and $\rho_r \geq (r-1)(r-2)/2 - 10 \geq r - 14$ for $r \geq 7$. □

Lemma 2.3 (Span bound)**.** *For a positive integer set X of size $r \geq 15$, with support hull-hole count h_X, one has*

$$\max X \leq r + \rho(X) + h_X + 2.$$

Proof. Delete the largest element $r+q$ and let $\nu \geq 0$ be the number of previous defects it destroys. If primes denote the preceding prefix,

$$\rho(X) = \rho(X') + q - h_X + h_{X'} - 2 + 2\nu.$$

Now $\rho(X') \geq 0$, $h_{X'} \geq 0$, and $\nu \geq 0$. □

3 One common lattice and one exact budget

Assume for now that $0 \in A$ and $104 \leq n = |A| \leq 115$. Integralize before using an integer inverse theorem. Form B, D as in the covering lemma and make two four-element peels $D \supset D_1 \supset E$. The elementary comparison of (1) and two copies of (2) gives

$$s := |\operatorname{FS}(E)| \leq 22d + 2\Delta(D) + 77 \leq 24d + 77. \tag{4}$$

If $|B| \leq 7$, D is one-sided and E has at least $n - 16 \geq 88$ nonzero elements of one sign. Then $s \geq 1 + T_{88} = 3917$, contradicting $s \leq 24 \cdot 115 + 77 = 2837$. Thus $|B| = 8$ and

$$d = n-8, \qquad e = |E| = n-16, \qquad E = P \cup \{0\} \cup (-Q), \quad a = |P|,\ b = |Q|, \quad m = a+b = n-17.$$

3.1 Ensuring both signs and an opposite pair

If E contains p pairs $\{x, -x\}$ and q singly occurring absolute values, then $e = 2p + q + 1$ and

$$|\operatorname{FS}(E)| \geq 1 + T_{p+q} + T_p = \frac{e^2 + 3 + q^2}{4}. \tag{5}$$

Indeed translate the support to replace negative elements by their positive magnitudes. For ordered distinct magnitudes b_i, of multiplicity $m_i \in \{1, 2\}$, adjoining the m_i copies of b_i increases the support by at least im_i. The residues of $0, b_1, \ldots, b_{i-1}$ modulo b_i are distinct, and each of these classes gains at least m_i points. The smallest possible sum of these increments assigns multiplicity two to the first p positions, proving (5).

Combining with (4), where $d = e + 8$, gives

$$q^2 \leq 96e + 1073 - e^2, \qquad q \equiv e - 1 \pmod 2.$$

For $88 \leq e \leq 99$ this forces $p \geq 23$ (see the table below). Hence $a, b \geq 23$, both peels really removed two elements of each sign, and zero is not a diagonal exception in D.

3.2 The exact slack identity

Write $\mu = \mu(P) + \mu(Q)$ and $\rho = \rho(P) + \rho(Q)$. Set

$$\begin{aligned}
\Lambda &= 1 + T_a + T_b, & \kappa &= s - \Lambda - \mu,\\
\theta &= \delta_0(P) + \delta_0(Q) + 8 - \Delta(D), & c &= 2|2D| + 18d + 85 - |\operatorname{FS}(D)|,\\
\ell_1 &= |\operatorname{FS}(D)| - |\operatorname{FS}(D_1)| - |R(D)| + 1,\\
\ell_2 &= |\operatorname{FS}(D_1)| - s - |R(D_1)| + 1,\\
g &= 4d - 10 - (|R(D)| - |R(D_1)|), & \mathcal{T} &= \kappa + \ell_1 + \ell_2 + g.
\end{aligned}$$

All these slacks are nonnegative. For κ, use $|U + V| \geq |U| + |V| - 1$; for g, at most $4d - 10$ pairs involve a deleted point. The opposite pair just proved removes zero from the count of possible diagonal exceptions. Direct cancellation gives the exact identity

$$\boxed{\rho + 2\theta + c + \mathcal{T} = \Omega := 22d + 97 - \Lambda.} \tag{6}$$

3.3 Core normalization and propagation to the peeled layers

We use the following sufficient form of Stanchescu's inverse theorem [3, Corollary 2]. If integer sets U, V, each of size at least two, satisfy

$$|U+V| < |U| + |V| + \min(|U|, |V|) - 3,$$

their hull cardinalities on a common difference lattice are at most $|U+V| - |V| + 1$ and $|U+V| - |U| + 1$, respectively. For every possible oriented (a, b) with $a, b \geq 23$, $a + b = m$ and $\Omega \geq 0$, the table below verifies

$$T_{\min(a,b)} - 2 - \Omega > 0. \tag{7}$$

Thus $\kappa \leq \Omega < \min(|\operatorname{FS}(P)|, |\operatorname{FS}(Q)|) - 3$, which is a stronger sufficient hypothesis. Apply the theorem to $U = \operatorname{FS}(P)$ and $V = -\operatorname{FS}(Q)$ and divide by the common step. Because both supports contain zero, P, Q, E are now integral. If h_P, h_Q and z denote their support hull-hole counts, respectively, then

$$h_P, h_Q \leq \kappa, \qquad z = h_P + h_Q - \kappa, \qquad 0 \leq z \leq \min(h_P, h_Q) \leq \kappa. \tag{8}$$

Moreover $h_X < |\operatorname{FS}(X)| - 1$ for either side.

Let $R_p = \max P$ and $R_q = \max Q$. Lemma 2.3 gives $R_p + R_q \leq m + \rho + \kappa + z + 4$. Since $|R(E)| \leq 2(R_p + R_q) + 1$ and the next four points create at most $4e + 6$ restricted sums, with $C_* = 2m + 4e + 14 - \Lambda$ we have

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\leq C_* + 2\rho - \mu + \kappa + 2z + \ell_2 \\ &\leq C_* + 3\Omega. \end{aligned} \tag{9}$$

The second inequality follows from $z \leq \kappa$ and $\rho + \kappa + \ell_2 \leq \Omega$. The table verifies that it is strictly negative. Any added point off the core lattice would give two disjoint copies of $\operatorname{FS}(E)$ in $\operatorname{FS}(D_1)$, a contradiction. Hence D_1 is integral. Similarly,

$$|\operatorname{FS}(D)| - 2|\operatorname{FS}(D_1)| \leq 4d - 10 + \ell_1 - \ell_2 - s \leq 4d - 10 + \Omega - \Lambda < 0, \tag{10}$$

so D is integral. No integrality of the outer eight points is assumed yet.

In the table, "paired minimum" is the lower bound on p from (5), "inverse" is the minimum in (7), "inner" is the maximum of $C_* + 3\Omega$, and "outer" is the minimum of $\Lambda - (4d - 10 + \Omega)$. All oriented pairs, including both reflections, are checked.

n	oriented pairs	paired minimum	inverse	inner	outer
115	1	36	1223	-1845	2033
114	10	34	981	-1721	1961
113	15	33	854	-1597	1889
112	18	32	770	-1477	1819
111	21	30	695	-1357	1749
110	22	29	643	-1241	1681
109	25	28	581	-1125	1613
108	26	27	535	-1013	1547
107	29	26	486	-901	1481
106	30	25	446	-793	1417
105	31	24	408	-685	1353
104	34	23	371	-581	1291

The two extreme translates of $\operatorname{FS}(E)$ inside $\operatorname{FS}(D_1)$ overlap by (9); their hulls cover the new hull, leaving at most $2z$ holes. If their displacement is u, their union has at least $s + u - z$ points: its hull has u more lattice points than the old hull and at most $2z$ holes. Comparing this to the exact definition of ℓ_2 gives $\eta_2 \leq z + \ell_2$, where η_2 counts holes of $R(D_1)$ in its integer hull. The same argument at the other peel, using (10), gives

$$w := \operatorname{holes}(\operatorname{FS}(D)) \leq 4z, \qquad \eta_1 \leq 2z + \ell_1, \qquad \eta_2 \leq z + \ell_2. \tag{11}$$

4 The four-extreme defect budget

Lemma 4.1. *Let $F = \{x_1 < \cdots < x_v\} \subset \mathbb{Z}$, $v \geq 2$, and add four integers $a_1 < a_2 < x_1 < x_v < b_2 < b_1$ to obtain D. If η_F, η_D count restricted-sum hull holes and $g_H = 4v+6-(|R(D)|-|R(F)|)$, then*

$$\eta_D + \eta_F + g_H \geq 2v - 4.$$

Proof. Put $L = x_1 + x_2$, $U = x_{v-1} + x_v$, $h_+ = b_1 + b_2 - U$, $h_- = L - a_1 - a_2$, and $h = h_+ + h_-$. Let W_+ be the union of the portions of $b_1 + F, b_2 + F$ above U. Write $u = b_2 - x_v$, $v_0 = b_1 - b_2$, $e_0 = x_v - x_{v-1}$, and $I_+ = (U, b_1 + x_v]_{\mathbb{Z}}$. If $H_+ = |I_+| - |W_+|$, then

$$|W_+| \leq u + 1 + v, \quad |W_+| = u + v_0 + e_0 - H_+, \quad 2|W_+| \leq h_+ + v + 1 - H_+.$$

For the first bound, the terms of $b_2 + F$ above U use at most u integers up to x_{v-1} and the one point x_v; the other translate has at most v terms. Reflection gives the corresponding W_-, H_-. Apart from W_+, W_- and the two new extreme sums, at most four mixed pairs $a_i + b_j$ create new values outside $[L, U]$. Let q_0 count those not already counted. They lie in the unused parts of I_+, I_-, so $q_0 \leq 4$ and $q_0 \leq H_+ + H_-$. Therefore

$$2N_{\text{out}} \leq h + 2v + 6 - (H_+ + H_-) + 2q_0 \leq h + 2v + 10.$$

If $N = |R(D)| - |R(F)|$, then $N \leq N_{\text{out}} + \eta_F$ and $\eta_D = h + \eta_F - N$. Substitution yields

$$\eta_D + \eta_F + g_H = h + 2\eta_F + 4v + 6 - 2N \geq 2v - 4.$$

□

Apply the lemma with $F = D_1$, $v = d - 4$. Equations (8) and (11) imply, for $G = 2d - 12$,

$$\boxed{\mathcal{T} \geq \kappa + G - 3z \geq G - 2z, \qquad \mathcal{T} \geq z, \qquad \mathcal{T} \geq t_0 := \lceil G/3 \rceil.} \tag{12}$$

If $z > 0$, both h_P, h_Q are positive but smaller than their support cardinalities minus one. Thus neither support is an arithmetic progression: step one has no holes and step at least two has too many. Stability and Lemma 2.2 now give

$$\mu(P) \geq a - 3, \quad \mu(Q) \geq b - 3, \quad \rho \geq m - 28. \tag{13}$$

5 Twenty-four effective hole witnesses

Only the hole threshold 14 is needed. We give its complete quantitative proof to avoid any unspecified large-size assumption.

Lemma 5.1 (Counted endpoint holes)**.** *If $i \geq 2e_0 + 5$ and $y_i = e_0 \geq 0$, then, up to positive scaling,*

$$X_i = [1, i + e_0]_{\mathbb{Z}} \setminus D_0, \quad |D_0| = e_0,$$

and its support is

$$\mathcal{S}(L, H) := [0, L]_{\mathbb{Z}} \setminus (H \cup (L - H)), \quad H \subseteq [1, 2e_0]_{\mathbb{Z}}, \quad |H| \leq e_0.$$

Proof. The equality case handles $e_0 = 0$. Otherwise (3) first gives $\delta \leq e_0$ deletions and maximum $i + \delta$. Remove the maximum. Lev's theorem [4, Theorem 1] states that for $B \subseteq [1, L_0]_{\mathbb{Z}}$, $|B| = s_0$, and $L_0 \leq 3s_0/2 - 2$,

$$[2L_0 - 2s_0 + 1, \ \sum B - (2L_0 - 2s_0 + 1)]_{\mathbb{Z}} \subseteq \mathrm{FS}(B).$$

Apply this with $s_0 = i - 1$, $L_0 = i + \delta - 1$; the density condition is $i \geq 2\delta + 5$. By symmetry its support has endpoint holes in $[1, 2\delta]$. Its total is at least $T_{i-1} > i + 5\delta$. Adding $i + \delta > 2\delta$ overlaps the central intervals and keeps precisely the same endpoint holes. Its support increment is $i + \delta$, so $\delta = e_0$. Every lower hole is absent from the coefficient set itself; there are only e_0 missing coefficients. Hence $|H| \leq e_0$. □

Lemma 5.2 (Effective hole test)**.** *For an integer $C \ge 0$, define*

$$K = \max\{k \ge 3 : k(k-3)/2 \le C+1\}, \qquad D_* = C + K, \qquad E_* = D_* + 1,$$
$$M = \max\left\{14, 1 + \left\lfloor \frac{1+\sqrt{9+8(E_*+42)}}{2} \right\rfloor\right\}, \qquad J = \max\{K+1, 19, M\}.$$

If

$$7(r - J + 1) > D_*, \qquad T_{J-1} - 1 > D_*, \tag{14}$$

a positive r-element set whose support has, on a specified lattice, $14 < h < |\,\mathrm{FS}\,| - 1$ must satisfy $\mu \ge r + C + 1$.

Proof. Assume $\mu \le r+C$ and divide by the specified lattice step. The set is integral and primitive: a gcd at least two would create at least $|\,\mathrm{FS}\,|-1$ holes. Its support is not a progression. If no prefix of size at least three has progression support, then $y_j \ge 1$ for $j \ge 4$ and $\sum_{j=4}^{r}(y_j-1) \le C+3 \le D_*$. Otherwise let $k \ge 3$ be the last progression prefix. The next support translate is disjoint, giving $y_{k+1} \ge T_k - k$. Subsequent increments are positive, so

$$\mu \ge r - 1 + k(k-3)/2, \qquad k \le K.$$

In either case a tail starting no later than $K+1$ satisfies

$$y_j \ge 1, \qquad \sum(y_j - 1) \le D_*. \tag{15}$$

The first test in (14) forces some $i \in [J, r]$ with $1 \le y_i \le 7$. Lemma 5.1 gives a scaled deleted interval with endpoint state $H \subseteq [1, 14]$, $|H| \le 7$. Its coefficient set has gcd one, so its scale t is a positive integer. If $t > 1$, primitivity of the full set gives a first later coefficient off $t\mathbb{Z}$. At that step the supports are disjoint and

$$y_j - 1 \ge T_{j-1} - j \ge T_{i-1} - 1 \ge T_{J-1} - 1 > D_*,$$

contradicting (15). Thus $t = 1$.

Every later excess is at most E_*. We verify that the endpoint state persists. For a current prefix of size $v \ge M$ with support $\mathcal{S}(L, H)$, $L \ge T_v$, and its next element $x > v \ge 14$. If $x \ge L - 28$, the portion of $x + \mathcal{S}(L, H)$ above L has at least $L - 42 \ge T_v - 42 > v + 1 + E_*$ points, contradicting its allowed increment. The last strict inequality is exactly the definition of M. Hence $14 < x < L - 28$. The two central integer intervals overlap or abut; the lower holes remain H and symmetry gives upper holes $L + x - H$. Induction leaves at most $2|H| \le 14$ holes in the final support, a contradiction. □

The following witnesses are sufficient, with no assertion of optimality. The last columns are $7(r-J+1)-D_*$ and $T_{J-1}-1-D_*$, respectively. The program also verifies $T_M - 42 > M+1+E_*$ using integer arithmetic.

r	C	K	D	J	window margin	off-scale margin
32	83	14	97	19	1	73
33	89	15	104	19	1	66
34	96	15	111	19	1	59
35	102	15	117	19	2	53
36	109	16	125	19	1	45
37	110	16	126	19	7	44
38	116	16	132	20	1	57
39	122	17	139	20	1	50
40	128	17	145	20	2	44
41	129	17	146	21	1	63
42	135	18	153	21	1	56
43	142	18	160	21	1	49
44	147	18	165	21	3	44
45	149	18	167	22	1	63
46	155	19	174	22	1	56
47	162	19	181	22	1	49
48	167	19	186	22	3	44
49	168	19	187	23	2	65
50	175	20	195	23	1	57
51	182	20	202	23	1	50
52	187	20	207	23	3	45
53	188	21	209	24	1	66
54	195	21	216	24	1	59
55	202	21	223	24	1	52

6 A covering certificate retaining the core masses

This section includes centers outside the target hull. Their errors are absorbed by the available source span, so no additional case enumeration for such centers is needed. Write the eight outer elements as

$$p_1 > p_2 > p_3 > p_4 > M_p := \max D, \qquad q_1 > q_2 > q_3 > q_4 > M_q := -\min D.$$

Let S_p, S_q be the sums of the positive and negative magnitudes in D. Put

$$\begin{gathered}\varepsilon_p = \sum P - T_a = \mu(P) + h_P, \qquad \varepsilon_q = \sum Q - T_b = \mu(Q) + h_Q,\\ F = S_p + S_q + c - 2(M_p + M_q) - 2(R_p + R_q),\\ F_A(a,b) = \tfrac{1}{2}T_a + T_b + 6a + 8b + 85,\\ F_B(a,b) = \tfrac{2}{3}T_a + T_b + 6a + \tfrac{22}{3}b + \tfrac{236}{3}.\end{gathered}$$

Every sum involving a core point, and every mixed-sign sum from $D\backslash E$, lies in $[-M_q-R_q, M_p+R_p]$. Only ten positive-positive and ten negative-negative pairs of the four peeled points per sign can lie outside. Thus $|2D| \le M_p + M_q + R_p + R_q + 21$. Using the definition of c and $|\,\mathrm{FS}(D)| = S_p + S_q + 1 - w$ gives

$$\boxed{F \le 18d + 126 + w \le 18d + 126 + 4z.} \tag{16}$$

Lemma 6.1 (All ordinary units active)**.** *If $c < d$, then after a possible reflection one of the following holds:*

$$F \ge F_A(a,b) + \tfrac{1}{2}\varepsilon_p + \varepsilon_q, \tag{17}$$

$$F \ge F_B(a,b) + \tfrac{2}{3}\varepsilon_p + \varepsilon_q. \tag{18}$$

There is no hypothesis limiting the number of out-of-hull centers.

Proof. Define, on the positive side,

$$\lambda_p = p_1 - p_2 - p_3 - p_4, \quad \alpha_p = p_2 - p_3 - p_4, \quad \beta_p = 2p_2 - p_1 - p_3 - p_4.$$

The ten ordinary centers are $\lambda_p + p_i$ $(1 \le i \le 4)$, $\lambda_p - q_j$ $(1 \le j \le 4)$, α_p, and β_p. Reflection defines the negative side, represented as positive targets and translates of $-D$. Pair $\lambda_p - q_1$ with $\lambda_q - p_1$. If $\beta_p + \beta_q \le 0$ (branch A), pair β_p with β_q; otherwise (branch B), pair $\lambda_p - q_2$ with $\lambda_q - p_2$. These are the two capacity-d pairings in (1).

If any of the eighteen units were empty, it would cost d in the cover slack, contrary to $c < d$. Every unpaired center therefore contributes to the actual support. The first pair's centers sum to less than $-(M_p + M_q)$, so at most one even meets its open half-line. In branch A the two unpaired centers $\lambda_p - q_2, \lambda_q - p_2$ both contribute, and their sum is greater than $-(M_p + M_q)$. The identity

$$\beta_p + \beta_q = -(\lambda_p - q_2 + \lambda_q - p_2) - 2(p_3 + p_4 + q_3 + q_4)$$

then makes $\beta_p + \beta_q < -(M_p + M_q)$. In branch B the second pair's sum is already less than $-(M_p + M_q)$ by the same identity. Thus each paired unit has precisely one contributing member.

All four own centers $b_i = \lambda_p + p_i$ contribute, and hence are integers because D and the target support are integral. Since $\lambda_p = -b_1 + b_2 + b_3 + b_4$, all p_i are integral. Reflection proves the same for all q_i.

Reflect so the first pair's contributor is $A_0 = \lambda_p - q_1$. In branch A put $B_0 = \lambda_q - p_2$, which contributes. Then

$$\beta_p = -(A_0 + B_0) - 2(p_3 + p_4) - q_2 - q_3 - q_4 < -M_p,$$

so the second contributor must be β_q. In branch B, β_p is unpaired and contributes, giving

$$p_2 - q_1 = A_0 + \beta_p + 2p_3 + 2p_4 > 2M_p.$$

It follows that $\lambda_q - p_2 < -M_q$, and the second contributor is $\lambda_p - q_2$. This proves that the following two combinations exhaust the actual configurations.

For each selected slot, let l_v, u_v count the points of its translate below 1 and above its target maximum S_p or S_q. Hull counts dominate actual target counts; the cover, with 84 fixed contributions, gives

$$\sum_v (l_v + u_v) \le c. \tag{19}$$

Define the nonnegative span allowances

$$U = M_p - a - 4, \quad V = M_q - b - 4, \quad q_p = R_p - a \le U, \quad q_q = R_q - b \le V.$$

A contributing positive center obeys

$$v \ge b + 5 - l_v - U\mathbf{1}_{v<0}, \qquad v \le S_p - a - 4 + u_v + V\mathbf{1}_{v>S_p}. \tag{20}$$

For $0 \le v \le S_p$, count the $b + 4$ negative elements and zero at the lower end, and the $a + 4$ positive elements at the upper end. If $v < 0$, contribution implies $-v < M_p$; among $[1, -v]$ at least $-v - U$ integers belong to the positive part of D. This proves the corrected lower bound. If $v > S_p$, contribution implies $v - S_p < M_q$; the corresponding count of negative magnitudes proves the upper bound. The negative-side versions interchange a, b and U, V.

In branch A use the lower centers

$$\lambda_p - q_1, \quad \lambda_q - p_2, \quad \lambda_q - p_3, \quad \alpha_p, \quad \beta_q$$

with coefficient one and the upper center $O = \lambda_p + p_1$ with coefficient $1/2$. Their exact difference is

$$\tfrac{1}{2}O - ((\lambda_p - q_1) + (\lambda_q - p_2) + (\lambda_q - p_3) + \alpha_p + \beta_q) = \tfrac{1}{2}p_2 + \tfrac{5}{2}p_3 + \tfrac{3}{2}p_4 + 3q_3 + 3q_4.$$

All losses used are distinct formal slots with coefficients at most one, so (19), (20), and integral outer spacings give

$$\tfrac{1}{2}S_p + c \geq \tfrac{9}{2}M_p + 6M_q + \tfrac{7}{2}a + 2b + 44 - \xi_A, \qquad \xi_A \leq 2U + \tfrac{7}{2}V. \tag{21}$$

The corrections count two positive lower rows, three negative lower rows, and half a positive upper row.

In branch B use lower centers $\lambda_p - q_1, \lambda_q - p_3, \lambda_q - p_4, \beta_q$ with coefficient one, β_p with coefficient $1/3$, and the upper center O with coefficient $1/3$. Here the difference is $2p_3 + 2p_4 + 3q_3 + 3q_4$, and the result is

$$\tfrac{1}{3}S_p + c \geq 4M_p + 6M_q + \tfrac{10}{3}a + \tfrac{4}{3}b + 38 - \xi_B, \qquad \xi_B \leq \tfrac{4}{3}U + \tfrac{10}{3}V. \tag{22}$$

Finally retain the exact core masses:

$$S_p \geq T_a + \varepsilon_p + 3R_p + 6 + M_p, \qquad S_q \geq T_b + \varepsilon_q + 3R_q + 6 + M_q.$$

Substitute these into the unused $S_p/2 + S_q$ in branch A and $2S_p/3 + S_q$ in branch B. After subtracting $2(M_p + M_q) + 2(R_p + R_q)$, the constant parts are exactly F_A, F_B. The residual span terms, after all corrections, are at least

$$\begin{aligned} 3U + 5V - \tfrac{1}{2}q_p + q_q - \xi_A &\geq \tfrac{1}{2}U + \tfrac{3}{2}V \geq 0, \\ \tfrac{8}{3}U + 5V + q_q - \xi_B &\geq \tfrac{4}{3}U + \tfrac{5}{3}V \geq 0. \end{aligned}$$

This proves (17) and (18), even when several centers are outside their target hulls. Formal-slot multiplicities are retained throughout, so coincidences between numerical centers do not invalidate the loss accounting. □

7 Exact exclusion of all remaining scalar states

There are finitely many necessary integer parameters, not finitely many candidate real sets. For each $104 \leq n \leq 115$, enumerate all oriented integers $a, b \geq 23$, $a + b = n - 17$, $\Omega \geq 0$. There are 262 such orientations in total. The preceding proofs apply to every actual set with these parameters.

7.1 At most fourteen core holes

For each $z \in \{0, \ldots, 14\}$ define

$$\rho_0 = \begin{cases} 0 & z = 0, \\ m - 28 & z > 0, \end{cases} \qquad c_{\max} = \Omega - \rho_0 - \max\{t_0, z, G - 2z\}.$$

If $c_{\max} < 0$, the state is impossible. Otherwise the program checks $c_{\max} < d$, licensing Lemma 6.1. For $z = 0$ use $\varepsilon_p, \varepsilon_q \geq 0$; for $z > 0$ use

$$\varepsilon_p \geq a - 3 + z, \qquad \varepsilon_q \geq b - 3 + z.$$

It then checks that both lower bounds in (17), (18) strictly exceed $18d + 126 + 4z$. All 416 surviving scalar states fail in this way. These are necessary relaxations; they need not be realizable.

7.2 At least fifteen core holes

If $z \geq 15$, first require $\Omega-(m-28) \geq t_0$, or the budget is already impossible. The 113 orientations surviving this condition have both side sizes in $[32, 55]$, exactly within the explicit witness table. For either side put

$$u(r) = r + C(r) + 1, \qquad v(r) = \max\{r - 14, u(r) + 2 - 2r\}.$$

Because $h_X \geq z > 14$ and $h_X < |\mathrm{FS}(X)| - 1$, Lemma 5.2 gives $\mu(X) \geq u(r)$, and $\delta_0(X) \leq r$ gives $\rho(X) \geq v(r)$. Let $H = \Omega - v(a) - v(b)$. Then

$$\mathcal{T} + c \leq H, \qquad z \leq \mathcal{T} \leq H, \qquad c \leq H - t_0.$$

If $H < t_0$, discard the state. Only six orientations remain. In every one, $H - t_0 < d$. Use $\varepsilon_p \geq u(a) + z$ and $\varepsilon_q \geq u(b) + z$ in the active certificate. The two margins over (16) are bounded below by

$$F_A(a,b) + \tfrac{1}{2}u(a) + u(b) - (18d + 126) - \tfrac{5}{2}H,$$
$$F_B(a,b) + \tfrac{2}{3}u(a) + u(b) - (18d + 126) - \tfrac{7}{3}H.$$

Both are strictly positive for every remaining orientation.

7.3 Results and the real-set conclusion

In the table below, a dash means that the preceding budget already discarded all states. A reported margin is the minimum of both certificate margins over all states in that column. All are exact rational numbers. The c columns give the largest remaining upper bound on c, which is strictly below $d = n - 8$.

n	low states	c_{low}	low margin	high states	c_{high}	high margin
115	0	–	–	0	–	–
114	0	–	–	0	–	–
113	0	–	–	0	–	–
112	0	–	–	0	–	–
111	0	–	–	0	–	–
110	0	–	–	0	–	–
109	0	–	–	0	–	–
108	0	–	–	0	–	–
107	9	18	$\frac{545}{2}$	0	–	–
106	42	43	209	0	–	–
105	151	68	168	0	–	–
104	214	92	$\frac{263}{2}$	6	7	351

This excludes all zero-containing feasible sets of sizes 104 through 115. Any feasible real set has size at most 115 by the audited foundation. If one of size at least 104 omitted zero, adjoining zero would either contradict that foundation or give one of the excluded zero-containing sizes. This proves Theorem 1.1. No step assumes that feasibility passes to arbitrary subsets.

8 Reproduction, audit boundary, and the next obstruction

The new verifier is `verification/upper103.py`, using Python 3.10+ and only the standard library. From the repository root run:

```
python verification/upper103.py \
  --output results/2026-10-14/upper103/result.json
```

It reconstructs the four center identities over the rationals, checks span absorption, all 24 effective-hole witnesses and all strict gates, all lattice margins, all low-hole states and all high-hole orientations. Integer square roots and rational arithmetic are used throughout. No floating-point tolerance, external solver, archived endpoint certificate, or claimed historical PASS result is an input. Checks use explicit exceptions and remain enabled under `python -O`.

The output is an arithmetic audit of the displayed reductions. It is not a formal proof-assistant certification of the analytic lemmas or of the cited theorems. Its two deliberately invalid effective-hole rows are rejected. It also records the obstruction at the next endpoint:

$$n = 103, \quad a = b = 43, \quad z = 0, \quad d = 95, \quad \Omega = 294, \quad G = 178, \quad c \le 116.$$

Here $116 \ge 95$, so the proof does not force all ordinary units active. This is a failure of the current relaxation, not a feasible 103-element example. Continuing below 103 requires a stronger slack bound or a proof covering at least one empty unit. The verifier deliberately does not certify this next endpoint with the all-active lemma.

The threshold-14, non-progression, effective-hole, and active-center ideas come from the archived continuation. This report rechecks and consolidates the needed portion into one uniform descent, retains the exact core mass, and makes the outside-center span absorption explicit. It leaves the longer 103-to-84 chain outside the current independent audit.

References

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